

What are DevTools?

DevTools can be defined as a 'suite of performance and debugging tools for Dart and Flutter.' It is presently under development.

One can perform several tasks using DevTools:

- Inspect the UI layout and the state of a Flutter app.
- Diagnose the UI jank performance issues in the Flutter application.
- Enables a source-level debugging and Dart command-line app.
- Debug issues related to memory in a Flutter and Dart command-line app.
- Enables viewing general and diagnostics information about a functioning Flutter or Dart command-line app.

As such, DevTools can be used with one's existing IDE or command-line based development workflow.

Upgrading Flutter

One can use the flutter command to upgrade one's Flutter SDK and the package that one's app depends on, irrespective of the Flutter release channel one follows.

Flutter basically comprises of four release channels: stable, beta, dev and master. If one is not using a very recent release, stable is the apt release channel to be used.

Hot Reload

Flutter has a hot reload feature which helps one rapidly experiment and build on user interfaces, fix bugs, and add new features. Hot reloads function by inserting updated source code files into the functioning Dart Visual Machine or VM. After the VM updates match with the new versions of fields and functions, widget trees are automatically rebuilt by the Flutter framework. Hence one can quickly view the effects of changes one has made.

When one invokes the hot reload feature, the host machine looks at the edited code since it was last compiled. Certain libraries are recompiled: the libraries with changed codes, the application's main library and the libraries which emerge from the main library creating affected libraries. Dart 2 turns the Dart source code of these libraries into kernel files. It then sends them to the mobile devices Dart VM. The Dart VM reloads all libraries from the new kernel file. No code is re-executed. After this, the hot reload mechanism causes Flutter to trigger a rebuild of the existing widgets and render objects.